

Refilling the Buffer in a Storm

The Netherlands' countercyclical capital buffer, 0% → 2% (2022–2024)

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Micro- & Macroprudential Management · Country implementation of a CCyB increase

Master of Financial Technology · Frankfurt School of Finance & Management

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SAY

- Good morning. We are Zheng Huang and Diego Gutiérrez, Master of Financial Technology.
- Our case for Micro- and Macroprudential Management is the Netherlands: how and why De Nederlandsche Bank, DNB, raised its countercyclical capital buffer from zero to two percent.

NUMBERS

- CCyB 0% → 2% (fully binding 31 May 2024)

TERMS EXPLAINED

- **DNB** — De Nederlandsche Bank, the Dutch central bank and macroprudential authority

The Netherlands raised its countercyclical capital buffer (CCyB) to 2% when the rulebook said 0% — refilling a buffer, not fighting a boom

SUMMARY

Decision: CCyB 0% → 1% (ann. May 2022) → **2%** (ann. May 2023, binding 31 May 2024), together with lower buffers for systemically important banks (O-SII).

Puzzle: the Basel reference indicator, the credit-to-GDP gap (credit above its long-run trend), stood at **−33pp** and **−47pp** at the two decisions → the buffer guide implied **0%**.

Q1 · Environment

Recovered economy, then war, energy shock and 11.6% inflation; ECB hikes from Jul-2022.

Credit cold; housing hot (+19% y/y), then falling.

Banks strong (core capital ratio, CET1, 17.7%) and more profitable as rates rose.

Q2 · Criteria & risks

DNB framework (2022): 2% = “**standard risk environment**”, calibrated on past crisis losses (€12bn), not on credit growth.

Insures against shocks no indicator predicts; housing left to other tools.

Keeps the March-2020 promise to restore releasable capital.

Q3 · Implications

Cheap: ≤ about one fifth of big banks' CET1 headroom.

A swap: structural buffers cut in 2020 and 2024.

No sign of contraction in credit or its price.

Complements ECB tightening; reciprocated abroad; **€6.7bn** releasable.

Verdict: **normalisation, not tightening.** **Trade-off:** relies on discretion and credibility; release never tested.

Source: DNB (2020, 2022, 2023, 2026); DNB Financial Stability Reports 2022–23; BIS credit-to-GDP gaps; ECB; bank disclosures; own calculations.

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2/24

SAY

- This slide is the whole talk; you can come back to it at any point.
- DNB raised the buffer twice, to two percent, while the Basel reference indicator, the credit-to-GDP gap, pointed to zero.
- The three boxes answer the three questions of the assignment: environment, criteria and risks, implications.
- Our verdict: **normalisation, not tightening.** DNB was refilling a buffer, not fighting a credit boom.
- The cost to banks was small, it was largely offset by cutting other buffers, and we see **no sign of a contraction in lending.**

NUMBERS

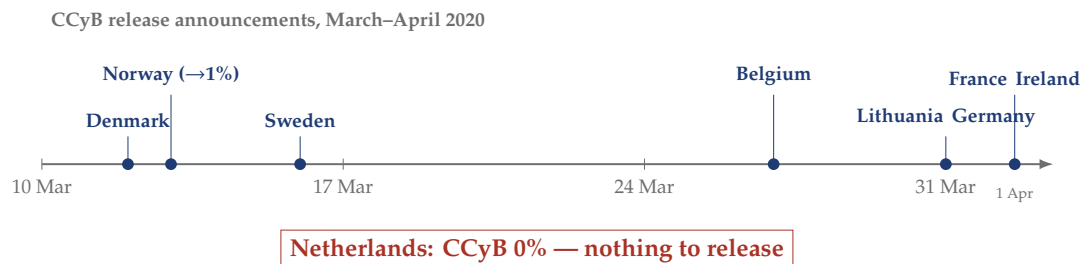
- Announcements: May 2022 (1%) and May 2023 (2%); binding 25 May 2023 and 31 May 2024
- Credit-to-GDP gap at the decisions: −33pp and −47pp → Basel guide 0%
- Inflation 11.6% (2022) · CET1 17.7% (end-2021) · €6.7bn releasable

TERMS EXPLAINED

- **CCyB** — countercyclical capital buffer: extra bank capital (0–2.5% of risk-weighted assets) built in good times, released in bad times
- **Credit-to-GDP gap** — how far credit/GDP is above its long-run trend; the Basel starting point for the CCyB
- **O-SII** — “other systemically important institution”; the O-SII buffer is an extra capital charge on large banks

When COVID hit, neighbours released their buffers — the Netherlands had nothing to release

PROLOGUE



DNB improvised (17 Mar 2020):

- cut systemic buffers of ING, Rabobank, ABN AMRO: 3% → 2.5 / 2 / 1.5%
- postponed a planned floor on mortgage risk weights
- €8bn freed, up to €200bn of lending

... and made a promise

"Once the situation is back to normal, DNB will compensate the systemic buffers reduction by gradually increasing the countercyclical capital buffer to 2%."

Source: ESRB, CCyB rates table (release announcements, Mar–Apr 2020); DNB press release "DNB lowers bank buffer requirements to support lending", 17 Mar 2020.

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3/24

SAY

- March 2020: COVID hits Europe. Within three weeks, a string of countries **release** their countercyclical buffers, telling banks to use that capital to keep lending.
- Sweden and Norway had been at 2.5 percent, the maximum normally used. Denmark, Ireland and Lithuania were at one percent. Germany and Belgium cancelled increases that had been announced but not yet applied.
- The Netherlands could not do this. Its buffer was at **zero**, so there was **nothing to release**.
- So DNB improvised. On 17 March 2020 it cut the systemic buffers of ING, Rabobank and ABN AMRO and postponed a planned floor on mortgage risk weights. Together that freed about eight billion euros.
- And it made a promise (*pause*): once things were back to normal, it would rebuild this capital as a two-percent countercyclical buffer. The whole story turns on that promise.

NUMBERS

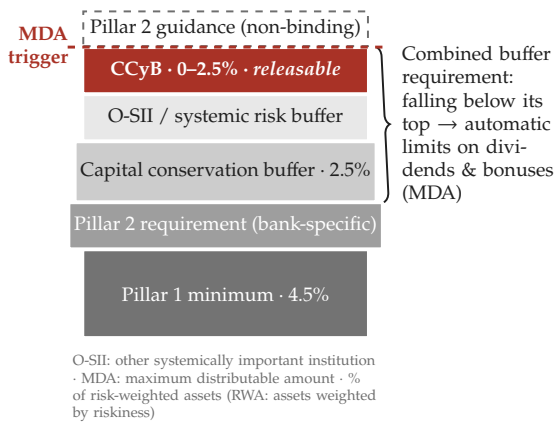
- Releases (rate in force → new rate):
 - Denmark 1.0% → 0% (12 Mar)
 - Norway 2.5% → 1.0% (13 Mar)
 - Sweden 2.5% → 0% (16 Mar)
 - Lithuania 1.0% → 0% (31 Mar)
 - France 0.25% → 0% (1 Apr)
 - Ireland 1.0% → 0% (1 Apr)
- Cancelled pending increases: Belgium 0.5% (27 Mar) · Germany 0.25% (31 Mar)
- Also: Iceland 2.0% → 0% · Czech Republic 1.75% → 1.0%
- NL: systemic buffers 3% → 2.5% (ING) / 2% (Rabo) / 1.5% (ABN AMRO); €8bn freed (incl. the floor postponement); up to €200bn of lending

TERMS EXPLAINED

- **Release** — the authority lowers the CCyB rate with immediate effect, so banks may use that capital to absorb losses and keep lending
- **Systemic (risk) buffer** — a structural capital charge on large banks; not designed to be released
- **Mortgage risk-weight floor** — a minimum average risk weight on Dutch mortgages for banks using internal models

The CCyB is the only layer of bank capital designed to be switched off in a crisis

PROLOGUE



Design

- 0–2.5% of risk-weighted assets, on *domestic* exposures
- Increases bind after 12 months; releases are immediate
- EU reciprocity: foreign banks lending in NL hold it too

Why “releasable” matters

- In COVID, banks close to the MDA trigger **cut lending** even when *allowed* to use buffers (stigma, payout limits)
- Only an explicitly **released** buffer is truly usable

Source: CRD (Arts. 130–136); BCBS (2024) d585; Couaillier, Lo Duca, Reghezza & Rodriguez d’Acri (2022), ECB WP 2644. Stack not to scale.

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4/24

SAY

- Not all capital is usable in a crisis. Read the stack from the bottom.
- First come **requirements** that must always be met: the legal minimum and the bank-specific Pillar 2 requirement.
- On top sits the **combined buffer**: the conservation buffer, the systemic buffer and the CCyB.
- The red dashed line is the MDA trigger. A bank whose capital falls below the top of the combined buffer automatically faces limits on dividends and bonuses. So banks avoid dipping into buffers, even when allowed to; markets punish it.
- The CCyB is the only layer the authority can **switch off**. When it is released, the requirement itself falls, and there is no stigma.
- In COVID, ECB research found that banks close to the MDA trigger cut lending anyway. Only explicitly released capital really works.

NUMBERS

- Pillar 1 minimum 4.5% CET1 · capital conservation buffer 2.5%
- CCyB 0–2.5% of RWA (higher possible) · increases bind after 12 months · releases are immediate

TERMS EXPLAINED

- **RWA** — risk-weighted assets: assets weighted by riskiness (a mortgage counts less than a corporate loan)
- **CET1** — Common Equity Tier 1: the highest-quality capital (shares, retained earnings), as % of RWA
- **MDA** — maximum distributable amount: the cap on payouts that applies once capital falls below the combined buffer requirement
- **Pillar 2 requirement / guidance** — the supervisor’s bank-specific add-on (binding) / extra expectation (non-binding)

NEXT → “Two years later DNB filled exactly that layer, and the timing looks strange.”

DNB raised capital requirements in the middle of a storm, when its own rulebook said zero. Wrong moment?

THE QUESTION

The storm (2022–23)

War in Ukraine · record energy prices
 HICP inflation **11.6%** (2022)
 ECB hikes from **21 Jul 2022**
 House prices turning down
 Basel gap **–33 / –47pp** → guide **0%**

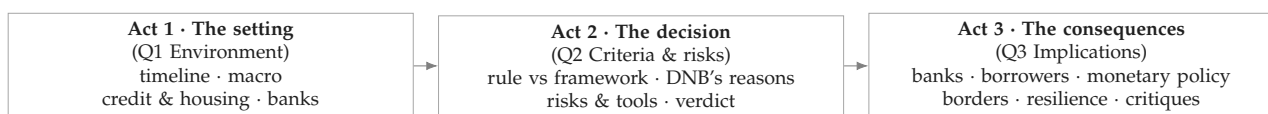


The decision

CCyB **0% → 1% → 2%**
 announced May 2022 and May 2023
 binding May 2023 and May 2024

Raising capital in a downturn is the **procyclical mistake** — amplifying the cycle — that the CCyB was invented to avoid.

Tightening into a slowdown? Fighting a hidden boom? Or something else?



Source: CBS (HICP 2022); ECB monetary policy decision, 21 Jul 2022; BIS credit-to-GDP gaps (latest data at each decision: 2021Q4, 2022Q4).

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5/24

SAY

- In 2022 DNB keeps its promise, but look at the timing on the left: war in Ukraine, record energy prices, 11.6 percent inflation, ECB rate hikes, house prices about to turn. And the Basel indicator says zero.
- Raising capital requirements in a downturn is *procyclical*: it amplifies the cycle. That is exactly the mistake the CCyB was invented to prevent.
- So our question is: **was this the wrong moment?**
- We answer in three acts, the three questions of the assignment. The boxes show what each act covers, without overlap: the setting, the decision, the consequences.

NUMBERS

- HICP inflation 11.6% (2022) · ECB first hike 21 Jul 2022 (+50bp)
- Credit-to-GDP gap **–33pp** (data to 2021Q4) and **–47pp** (data to 2022Q4)

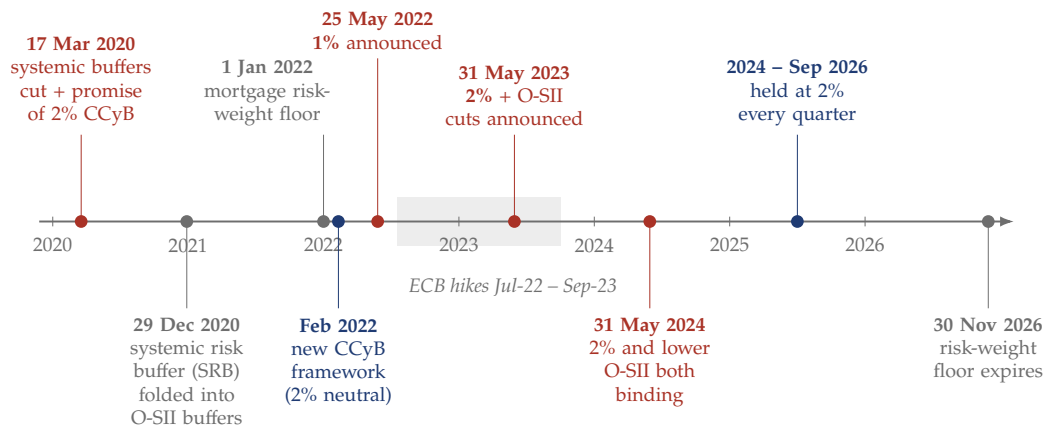
TERMS EXPLAINED

- **Procyclical** — policy that reinforces the cycle (tighter in bad times, looser in good times)
- **Basel buffer guide** — gap below 2pp → 0% buffer; above 10pp → 2.5%; linear in between

NEXT → "Act 1, the setting. First, the timeline."

The path to 2% was announced in 2020 — and walked step by step as the ECB began to tighten

Q1 · ENVIRONMENT



Source: DNB press releases (17 Mar 2020; 27 May 2022; 31 May 2023; 17 Sep 2026; 24 Apr 2026); ESRB O-SII notifications (27 Nov 2020; 31 May 2023); ECB (21 Jul 2022; 14 Sep 2023).

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6/24

SAY

- The path was set in March 2020 with the promise.
- In December 2020 the systemic risk buffer was folded into O-SII buffers under new EU rules, and in January 2022 the mortgage risk-weight floor finally came into force.
- In February 2022 DNB published its new CCyB framework. In May 2022 it announced one percent, just before the ECB began to hike in July. In May 2023 it announced two percent, together with lower O-SII buffers.
- Both changes became binding on the same day, 31 May 2024, and DNB has held the buffer at two percent every quarter since.
- Looking ahead: the mortgage floor expires at the end of November 2026.

NUMBERS

- 17 Mar 2020 · 29 Dec 2020 · 1 Jan 2022 · Feb 2022 · 25 May 2022 · 31 May 2023 · 31 May 2024 · 30 Nov 2026
- ECB hikes: first on 21 Jul 2022; still hiking on 14 Sep 2023

TERMS EXPLAINED

- **Announcement vs binding date** — banks get 12 months between a CCyB increase being announced and having to meet it
- **CRD V** — the 2019 revision of the EU Capital Requirements Directive; it made the O-SII and systemic risk buffers additive, prompting DNB's conversion

The economy had fully recovered — then war, energy and inflation darkened the outlook

Q1 · ENVIRONMENT

2021Q3

GDP back above
pre-COVID level

11.6%

HICP inflation, 2022
(national CPI 10.0%)

+50bp

ECB's first hike
21 July 2022

record

energy & commodity prices
after the invasion

DNB, Financial Stability Report, spring 2022

"After a strong recovery from the pandemic, the economic outlook has worsened due to the war in Ukraine and high inflation, and financial stability risks have increased."

"Higher funding costs may put pressure on the debt sustainability of governments, businesses and households."

By end-2021 the Dutch economy was almost 3% larger than at end-2019 — the recovery DNB's framework requires before building the buffer.

Source: CBS (GDP, Nov 2021; inflation 2022); ECB monetary policy decision 21 Jul 2022; DNB Financial Stability Report, spring 2022.

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7/24

SAY

- By the third quarter of 2021, Dutch GDP was back above its pre-COVID level; by the end of 2021 it was almost three percent larger than at the end of 2019. That recovery is the condition DNB's framework requires before building the buffer.
- Then the storm. DNB's spring 2022 Financial Stability Report: "the economic outlook has worsened due to the war in Ukraine and high inflation."
- Energy prices hit records, inflation reached 11.6 percent, and on 21 July 2022 the ECB raised rates by fifty basis points, its first hike.
- DNB also warned that higher funding costs would weigh on the debt sustainability of governments, firms and households.
- So the macro picture argued both ways: build now while the economy can take it, or wait because it is weakening.

NUMBERS

- GDP above pre-COVID level: 2021Q3 (+1.9% q/q) · end-2021 ≈ +3% vs end-2019
- HICP 11.6% in 2022 (national CPI 10.0%)
- ECB +50bp on 21 Jul 2022

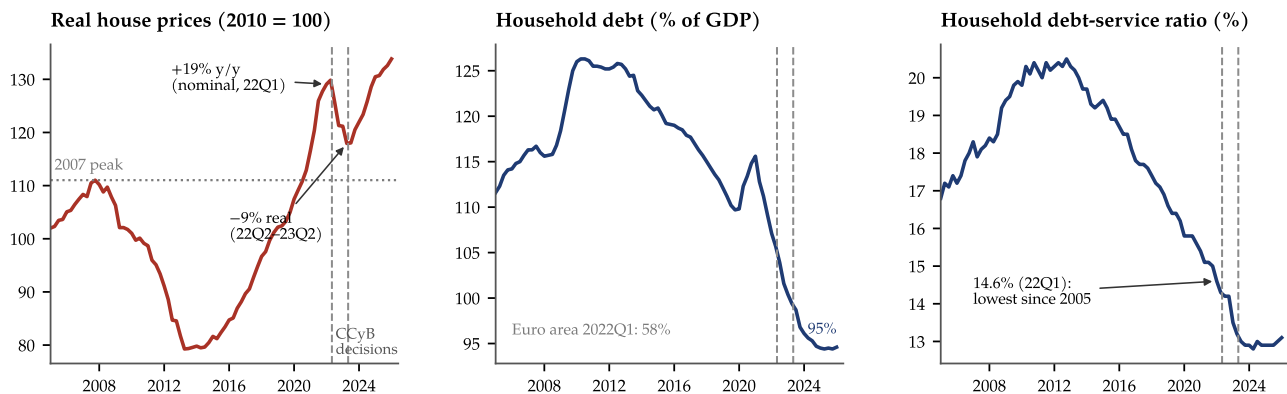
TERMS EXPLAINED

- **HICP** — Harmonised Index of Consumer Prices, the euro-area-comparable inflation measure
- **Basis point (bp)** — 0.01 percentage point; 50bp = 0.5pp

NEXT → "In the financial system the picture was split."

Credit was cold; housing was the one hot spot — and it turned

Q1 · ENVIRONMENT



Housing: nominal prices +19% y/y (22Q1), real 16% above the 2007 peak. DNB itself: “the overheated housing market calls for further measures.”

Credit: household debt high (107% of GDP vs 58% in the euro area) but falling; debt service at its lowest since 2005.

Source: BIS (WS_SPP, WS_TC, WS_DSR); DNB Financial Stability Report, spring 2022; own calculations. Gold lines: CCyB announcements (May-22, May-23).

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8/24

SAY

- Left panel, **housing was hot**. In early 2022 nominal house prices rose 19 percent year on year, and real prices were 16 percent above their 2007 peak. DNB itself called the housing market “overheated”.
- But watch what happens next: as mortgage rates rose, real prices fell by about nine percent within a year.
- Middle panel, **credit was cold**. Household debt is high by international standards, about 107 percent of GDP versus 58 percent in the euro area, but it was falling.
- Right panel: the household debt-service ratio, the share of income spent on interest and repayments, was at its lowest since 2005.
- If there was a boom, it was in house prices, not in credit.

NUMBERS

- Nominal house prices +19.0% y/y (2022Q1); real index 129 vs 2007 peak 111 (+16%)
- Real house prices –9.2% (2022Q2–2023Q2)
- Household debt/GDP: 110% (2019Q4) → 107% (2022Q1) → 100% (2023Q1); euro area 57.7% (2022Q1)
- Debt-service ratio, households: 14.6% (2022Q1)

TERMS EXPLAINED

- **Debt-service ratio** — interest plus principal payments as a share of income
- **Real house prices** — nominal prices deflated by consumer prices

NEXT → “Who would carry the losses if this went wrong? The banks.”

Banks entered the storm well capitalised — and rising rates were lifting their profits

Q1 · ENVIRONMENT

17.7%

core capital (CET1) ratio
end-2021 (EU 15.7%, 21Q3)

16.3%

CET1 ratio, end-2022
"in line with EU average"

+6.7%

net interest income 2022
(67.7% of income)

11.5%

4 major banks' CET1 after a
severe stress test (min. 8%)

Q1 in one line — against raising now

war and energy shock · 11.6% inflation
rapid ECB tightening · house prices turning

DNB (FSR 2022): higher rates "may ultimately have a positive impact on the profitability of financial institutions."

... and for raising now

economy recovered · credit cold
banks strong, profits rising with rates

Source: DNB Financial Stability Reports, spring 2022 and spring 2023.

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9/24

SAY

- The banks were in good shape. The core capital ratio was 17.7 percent at the end of 2021, above the EU average, and 16.3 percent a year later.
- In DNB's 2023 stress test the four major banks would lose 3.8 points of capital and still end at 11.5 percent, above the 8 percent minimum.
- And rising rates were lifting profits: net interest income grew 6.7 percent in 2022. DNB wrote that higher rates "may ultimately have a positive impact on the profitability of financial institutions". Keep that sentence in mind; it is our twist later.
- The bottom row is Act 1 in one line: against raising now, war, inflation, rate hikes and a turning housing market. For raising now, a recovered economy, cold credit and strong, increasingly profitable banks.

NUMBERS

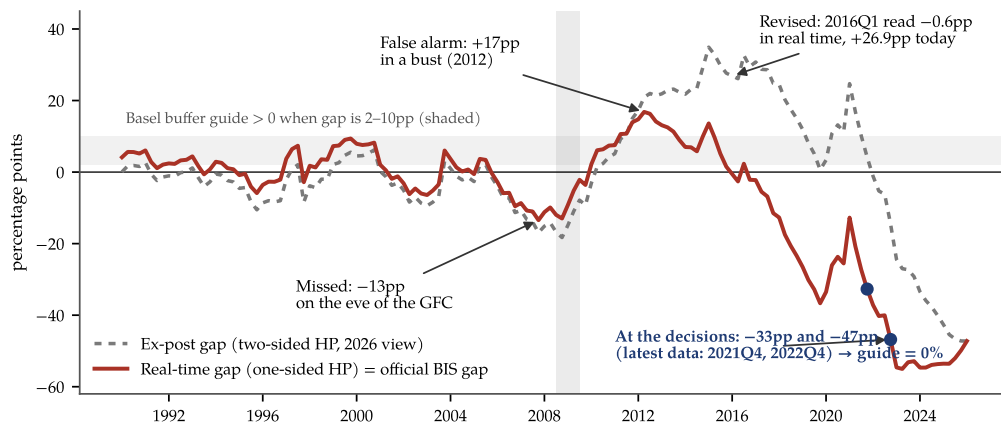
- CET1 17.7% (end-2021; EU 15.7% in 2021Q3) → 16.3% (end-2022, in line with EU)
- Stress test (FSR 2023): −3.8pp → 11.5% at end-2025 (min. 8%)
- Net interest income +6.7% in 2022; 67.7% of bank income

TERMS EXPLAINED

- **Stress test** — a simulation of a severe recession to check whether capital stays above the minimum
- **Net interest income** — interest earned on loans minus interest paid on deposits and funding

The Basel rule said 0% — but on Dutch data it had missed the last crisis and raised false alarms

Q2 · CRITERIA & RISKS



Credit-to-GDP gap = credit/GDP ratio minus its long-run trend (Hodrick-Prescott filter). **Rule:** gap < 2pp → 0%, gap > 10pp → 2.5%. Our one-sided filter ($\lambda = 400,000$) reproduces the official BIS gap within 0.00005pp (1971–2026).

Alternative trend filters are noisy around COVID (two-sided HP +4.2pp in 2021Q4) — one more reason not to trigger on any of them.

Source: BIS credit-to-GDP gaps (Q.NL.P.A); Drehmann et al. (2010); Repullo & Saurina (2011); Edge & Meisenzahl (2011); own calculations.

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10/24

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- Thank you. Let me start with the rule DNB chose *not* to follow.
- The red line is the credit-to-GDP gap, credit relative to its long-run trend. The trend is estimated with a statistical smoother, the Hodrick-Prescott filter. Our own calculation matches the official BIS series almost exactly.
- On Dutch data, the rule failed three times. It **missed** the financial crisis: minus 13 points in 2007. It raised a **false alarm** in 2012, during a house-price bust, because falling GDP inflated the ratio. And it gets **revised**: 2016 read slightly negative in real time but plus 27 with today's data.
- Why so bad? Dutch credit rose from under 50 percent of GDP in the 1960s to around 350 percent at its 2015 peak. The filter treats that structural deepening as trend, so once the ratio falls the gap turns deeply negative.
- At the two decisions it read minus 33 and minus 47 points, so the rule said zero. DNB had good reason to use a different compass.

NUMBERS

- Gap: −13.4pp (2007Q4) · +16.8pp (2012Q2) · 2016Q1 −0.6pp real time vs +26.9pp ex post
- −32.7pp (2021Q4) and −46.8pp (2022Q4) at the decisions
- Credit/GDP 47% (1961) → 352% (2015Q1 peak)
- Replication error ≤ 0.00005 pp (1971–2026)

TERMS EXPLAINED

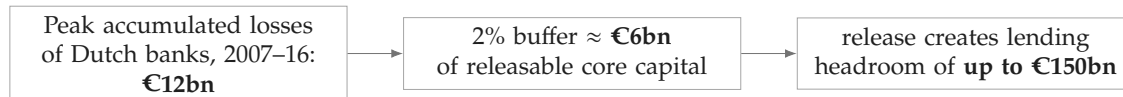
- **HP filter** — splits a series into a smooth trend and a cycle; $\lambda = 400,000$ for credit cycles
- **One-sided vs two-sided filter** — one-sided uses only data up to each date (what policymakers see); two-sided also uses later data (hindsight)
- **False alarm / missed crisis** — type II / type I errors of an early-warning indicator

DNB's compass: 2% is the normal-times level, calibrated on past crisis losses — not on credit growth

Q2 · CRITERIA & RISKS



built up 1pp a year → 2% after two years



- Criteria: a dashboard (macro, banks, households & firms, markets); “no mechanical link . . . guided discretion.”
- 2% chosen “also taking into account the impact of the buffer reduction in March 2020.”
- Not alone: 17 jurisdictions run a *positive neutral* CCyB (kept filled in normal times); NL, Sweden, UK, Poland target 2%. ECB loss-based range: 1.1–1.8%.

Source: DNB (2022), Analytical framework for setting the CCyB in the Netherlands; BCBS (2024) d585, Table 1; De Nora, Pereira, Pirovano & Stammwitz (2025), ECB WP 3061.

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11/24

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- That compass is DNB's 2022 framework, with four phases.
 - After a crisis, the buffer is released.
 - In normal times it should be at two percent, built up one point a year.
 - Only when risks are clearly elevated does it go above two percent.
 - In a crisis it is released.
- This is called a *positive neutral* buffer: already filled when risks are neither high nor low.
- Two percent is calibrated on history: Dutch banks' peak accumulated losses in past crises were about 12 billion euros. Two percent equals about six billion of releasable capital, and releasing it could support up to 150 billion of lending.
- DNB decides by *guided discretion*: a dashboard of indicators informs the decision but does not dictate it. It also chose two percent “taking into account the buffer reduction in March 2020”.
- The Netherlands is not alone: seventeen jurisdictions run such a framework; Sweden, the UK and Poland also target two percent.

NUMBERS

- Peak accumulated losses €12bn (2007–2016) → 2% ≈ €6bn → up to €150bn of lending
- Build-up: 1pp per year → 2% after two years
- 17 jurisdictions with a positive neutral CCyB (BCBS 2024); ECB loss-based range 1.1–1.8%

TERMS EXPLAINED

- **Positive neutral CCyB** — a CCyB kept above zero in a “standard” risk environment
- **Peak accumulated losses** — the largest cumulative loss a bank suffered over a crisis period
- **Guided discretion** — decisions informed by a published set of indicators, with judgement, not a formula

NEXT → “What reasons did DNB actually give?”

DNB justified the steps by recovery, uncertainty and its 2020 promise — not by credit growth

Q2 · CRITERIA & RISKS

May 2022 · 0% → 1%

"When the systemic risk buffers . . . were lowered in March 2020, we also announced our intention to **restore the buffers** by raising the CCyB."

"The risk profile is currently dominated by uncertainty caused by the war in Ukraine, but . . . the robust economic recovery . . . provides grounds for a gradual build-up."

Cyclical risks: "normal" to elevated.

May 2023 · 1% → 2%

"Although the credit-to-GDP gap shows **no signs of excessive credit growth**, some other indicators already point to higher cyclical risk."

Cited: investors' rising risk appetite · real-estate prices falling · debt sustainability of firms and governments · banks "robust, partly due to rising interest rates".

Both decisions: **a normal-to-elevated risk picture + strong banks + the promise**. Credit growth is never the reason.

Source: DNB Financial Stability Report, spring 2022; DNB press releases of 27 May 2022 and 31 May 2023.

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12/24

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- In 2022, DNB wrote: "When the systemic risk buffers were lowered in March 2020, we also announced our intention to restore the buffers by raising the CCyB." It cited the strong recovery, described risks as "normal to elevated", and acknowledged the uncertainty of the war.
- In 2023, it said explicitly that the credit gap shows "no signs of excessive credit growth". The reasons it cited: investors' rising risk appetite, falling real-estate prices, debt sustainability of firms and governments, and banks made more robust by higher rates.
- In neither decision was credit growth the reason. Both rest on a normal-to-elevated risk picture, strong banks and the promise.

NUMBERS

- 1% step: about €3.3bn of extra CET1 (announced 25/27 May 2022)
- 2% step: about €3.4bn (announced 31 May 2023)

TERMS EXPLAINED

- **Cyclical systemic risk** — risk that builds up over the financial cycle (credit, asset prices, risk-taking) and can materialise system-wide

NEXT → "So which risks was the buffer meant to cover?"

The CCyB insures against shocks nobody can forecast; the known hot spots got their own tools

Q2 · CRITERIA & RISKS

Financial-stability risks named by DNB (FSR 2022/23)	Instrument
War, energy and supply shocks · rapid tightening of financial conditions and market corrections · debt sustainability of governments, firms, households · bank turmoil abroad (SVB, Credit Suisse, 2023)	CCyB — releasable for <i>any</i> shock
“Overheated” housing · high mortgage debt of first-time buyers · interest-only loans	borrower limits (loan-to-value, loan-to-income), tax reform, mortgage risk-weight floor (Jan 2022 – Nov 2026)
Size and concentration of the large banks (sector 400% → 280% of GDP)	buffer for systemically important banks (O-SII , cut 2024)

Principle: one instrument per target (Tinbergen, 1952)

DNB on the 2020 recomposition: more releasable capital is “a valuable addition given the sensitivity/volatility of the Dutch economy to external events.” A releasable buffer is the only tool that works whatever the source of the shock.

Source: DNB Financial Stability Reports, spring 2022 and spring 2023; ESRB O-SII notification NL (31 May 2023); DNB (24 Apr 2026); Tinbergen (1952).

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13/24

SAY

- DNB's reports name the financial-stability risks, and each was matched to a tool.
- **Shocks nobody can forecast** go to the CCyB, because it can be released whatever the source: the war, an energy shock, a sudden tightening of financial conditions, debt-sustainability problems, or bank turmoil abroad such as SVB and Credit Suisse in 2023.
- **The known hot spot, housing**, went to other tools: limits on loan-to-value and loan-to-income, tax reform, and the mortgage risk-weight floor.
- **The size of the big banks** goes to the O-SII buffer. The sector shrank from about 400 to 280 percent of GDP, which is why that buffer was cut.
- This is the Tinbergen principle: one instrument per target. DNB itself said more releasable capital is valuable “given the sensitivity of the Dutch economy to external events”.

NUMBERS

- Banking sector: $\approx 400\%$ of GDP (when O-SII buffers were phased in) → 280% (end-2022)
- Risk-weight floor in force Jan 2022 – Nov 2026

TERMS EXPLAINED

- **LTV / LTI** — loan-to-value / loan-to-income: caps on mortgage size relative to house value or income
- **Tinbergen principle** — reach each policy target with its own instrument
- **Borrower-based measures** — rules on borrowers (LTV, LTI) rather than on bank capital

NEXT → “Now we can test our two suspects.”

The verdict: every prediction fits normalisation — none fits overheating

Q2 · CRITERIA & RISKS

If DNB were ...	normalising	fighting overheating	what happened	
Responding to credit indicators	no link	rises with them	gap, debt service, household debt ratio all <i>falling</i> ; CCyB <i>up</i>	✓
Path	+1pp a year, stop at 2%	follows risk	1% → 2% → held	✓
2023: house prices falling, rates jumping	raise anyway	pause	raised	✓
2024–26: house prices rising again (+10% y/y, Mar-26)	stay at 2%	go above 2% (framework phase 3)	held at 2%	✓
Other buffers	cut structural ones to compensate	keep them	cut in 2020 and 2024	✓

Five predictions, five matches: **normalisation**. The last row is the key to the whole story.

Source: BIS (WS_CREDIT_GAP, WS_DSR, WS_TC, WS_SPP); DNB press releases 2022–2026 (house prices: DNB, 26 Mar 2026); DNB framework (2022); ESRB O-SII notifications.

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14/24

SAY

- Was DNB fighting overheating or normalising? Each makes different predictions. Row by row:
 - Indicators.** Overheating says the buffer rises with them. In fact they all fell while the buffer went up.
 - Path.** Normalisation says one point a year, stop at two. That is exactly what happened.
 - 2023.** House prices were falling and rates jumping. An overheating fighter would pause; DNB raised anyway.
 - 2024–26.** House prices rose again, about ten percent a year by early 2026. An overheating fighter would go above two percent; DNB held at two.
 - Other buffers.** Normalising means cutting structural buffers to compensate. DNB did, twice.
- (*pause*) Five predictions, five matches.
- The strongest evidence against us is house prices at plus 19 percent in 2022. But DNB used other tools for housing, and when house prices fell, it raised the buffer anyway.
- That last row is the key to the whole story.

NUMBERS

- Gap, debt-service ratio and household debt ratio all falling in 2022–23
- House prices +10% y/y (DNB, Mar 2026) · CCyB held at 2% since May 2024

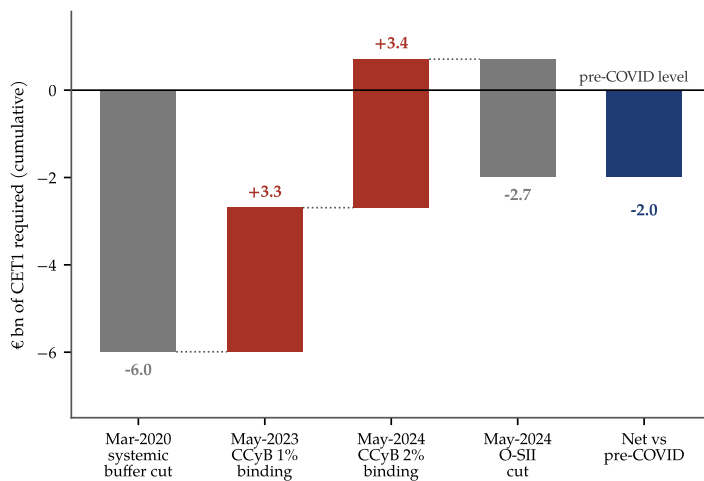
TERMS EXPLAINED

- Phase 3 ("increased risk")** — the framework phase in which the CCyB goes above 2%

NEXT → "Here is what that last row means in euros."

A swap, not a squeeze: DNB moved capital from structural to releasable buffers

Q2 · CRITERIA & RISKS



DNB's own words

"more or less capital-neutral for the three large banks" — 2020

"a limited increase in the net capital requirements" — Knot, Jun 2023

"slightly increases the capital requirements for the banking sector as a whole" — FSR 2023

Not unusual: the UK offset its 2% neutral CCyB by lowering Pillar 2A and resolution requirements.

Waterfall: approximate, own calculation. Buffer cuts valued at end-2022 total RWA of ING, Rabobank, ABN AMRO, de Volksbank (BNG excluded); CCyB amounts are DNB's sector-wide figures on Dutch exposures. Relative to 2021 the change was an increase.

Source: DNB press release 17 Mar 2020; Knot (2023), statement to the Standing Committee for Finance, 7 Jun 2023; DNB FSR spring 2023; BCBS (2024) d585; bank annual reports 2022.

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15/24

SAY

- This waterfall is our own approximate calculation.
 - In 2020 the systemic-buffer cut released about six billion euros of required capital.
 - The CCyB then added 3.3 billion in 2023 and 3.4 billion in 2024.
 - In 2024 the O-SII cut released another 2.7 billion.
- Net, relative to before COVID: roughly unchanged, if anything slightly lower.
- DNB said the same: the 2020 plan would be "more or less capital-neutral", and in 2023 Governor Knot spoke of "a limited increase in the net capital requirements".
- Two caveats. The bases differ: the cuts apply to big banks' worldwide assets, the CCyB to all banks' Dutch exposures. And measured from 2021, not 2019, there was a modest increase.
- So capital was **moved**, from buffers that cannot be released into one that can. The UK did something similar.

NUMBERS

- -6.0 (2020: ING 1.7, Rabo 2.4, ABN 1.9) · $+3.3$ (May-23) · $+3.4$ (May-24) · -2.7 (May-24 O-SII) → net ≈ -2.0 €bn vs pre-COVID
- Valued at end-2022 total RWA; BNG excluded

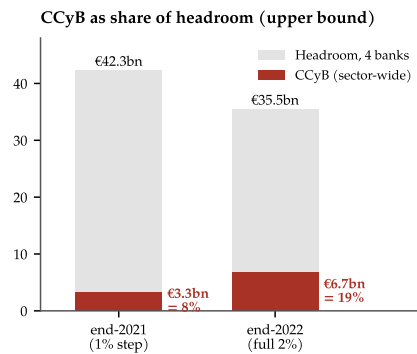
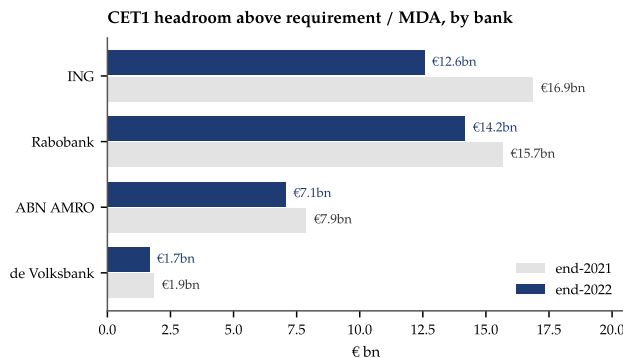
TERMS EXPLAINED

- **Structural buffer** — a capital charge for permanent features (size, interconnectedness), kept through the cycle
- **UK approach** — the UK offset its 2% neutral CCyB by lowering Pillar 2A and resolution requirements

NEXT → "But even a swap has a cost, especially in a storm. So, was it the wrong moment?"

The twist: the storm made it the cheapest moment — higher rates gave banks the profits to build the buffer

Q3 · IMPLICATIONS



Headroom = CET1 capital above the bank's requirement (MDA trigger). Net interest income +6.7% in 2022 → buffers built from *retained earnings*, not by cutting loans. Headroom shrank because of RWA growth and payouts, not the CCyB.

ECB: gradual build-up + high profitability "limit these economic costs" (MPB 24); activating buffers during the 2022–23 tightening "can mitigate implementation costs" (MPB 31).

Not the wrong moment — arguably the right one.

Source: Bank disclosures (ING, Rabobank, ABN AMRO, de Volksbank, 2021–22), de Volksbank requirement derived; DNB FSR 2023; ECB Macroprudential Bulletin 24 (2024) and 31 (2025, Box 2).

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16/24

SAY

- Here is the twist: the storm made it the **cheapest** moment.
- Headroom is the capital a bank holds above its requirement. The left panel shows it by bank: ING and Rabobank each had over ten billion euros, ABN AMRO about seven.
- In total the four banks had about 36 billion at the end of 2022. The full two-percent buffer is 6.7 billion, at most about a fifth, and that is an upper bound.
- Why so cheap? Rising rates lifted net interest income, so banks could build the buffer from retained earnings instead of cutting loans. Headroom shrank between 2021 and 2022 because risk-weighted assets grew and banks paid out capital, not because of the buffer.
- The ECB agrees: a gradual build-up with profitable banks "limits these economic costs", and activating buffers during the 2022–23 tightening "can mitigate implementation costs".
- Not the wrong moment. Arguably the right one.

NUMBERS

- Headroom end-2021 / end-2022 (€bn):
 - ING 16.9 / 12.6
 - Rabobank 15.7 / 14.2
 - ABN AMRO 7.9 / 7.1
 - de Volksbank 1.9 / 1.7
- Total headroom €42.3bn / €35.5bn
→ CCyB €3.3bn = 8%, €6.7bn = 19%

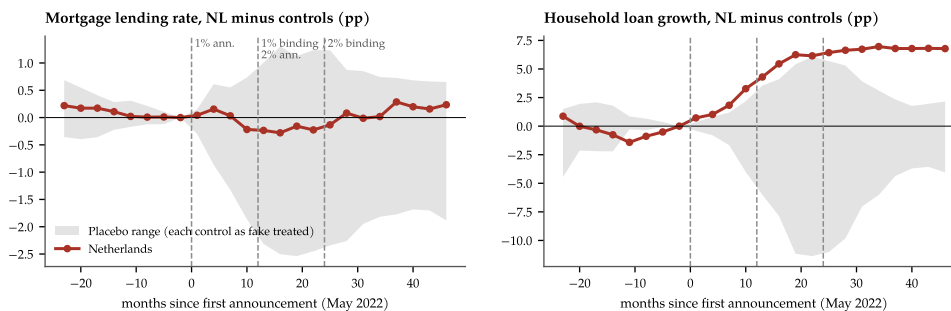
TERMS EXPLAINED

- **Headroom** — CET1 capital above the bank's requirement (its MDA trigger)
- **Retained earnings** — profits kept in the bank instead of being paid out

NEXT → "Cheap for banks, but did borrowers pay?"

Credit kept flowing — and its price moved with the rest of the euro area

Q3 · IMPLICATIONS



How to read: red = Netherlands minus countries that did not change their CCyB (AT, FI, IT, MT, LU); 0 = no difference; grey = differences that arise by chance (each control treated as if it had raised its CCyB).

- Dutch lending rates stayed **inside the placebo range** at all three CCyB dates
- Household credit grew **faster** than in peers (partly catch-up: NL lagged in 2019–22)

DNB (Sep 2026): “no signs of constraints in bank credit supply.”

One treated country, small control group, housing rebound: no sign of contraction, not a causal estimate.

Source: ECB Data Portal (MIR, BSI); ESRB CCyB table (control selection); BIS (WS_TC); DNB press release 17 Sep 2026; own two-way fixed-effects event study.

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17/24

SAY

- We ran an event study: Dutch lending rates and loan growth compared with five euro-area countries that never changed their buffer (Austria, Finland, Italy, Malta, Luxembourg), around each policy date.
- How to read it: the red line is the Netherlands minus those countries; zero means no difference. The grey band shows how big differences get by pure chance. We pretend each control country raised its buffer and plot those "fake" effects.
- Left: Dutch mortgage rates stay inside the band at all three dates. Right: household credit actually grew faster.
- Honest detail: before 2022 the Netherlands had lagged, so part of that is catch-up.
- Our claim is modest. With one treated country this is "no sign of contraction", not a causal estimate. DNB's own verdict in September 2026: "no signs of constraints in bank credit supply".

NUMBERS

- Dates: May-22 (1% announced) · May-23 (1% binding, 2% announced) · May-24 (2% binding)
- Bank credit 2022Q1–2026Q1: NL +16.6% vs euro area +7.9%; 2019–22: NL +4.3% vs +11.3%

TERMS EXPLAINED

- **Event study** — tracks the difference between treated and control groups before and after an event
- **Placebo test** — rerun the analysis pretending an untreated country was treated, to see what chance alone produces

Building the buffer while the ECB hiked did not double the squeeze — it let monetary policy focus on inflation

Q3 · IMPLICATIONS

The concern

Two tightenings at once: ECB rates *and* capital requirements → credit crunch?

What the data show

Dutch lending rates did not diverge from euro-area peers (event study); bank profits *rose* with rates (net interest income +6.7%).

ECB Macprudential Bulletin 31 (2025)

Early CCyB activation “helps monetary policy focus on its primary objective of price stability, thereby largely eliminating the potential for conflict between price stability and financial stability.”

DNB, FSR 2022

“A sudden tightening of financial conditions could also have a negative impact on financial stability.”

One monetary policy for the euro area, but national macroprudential policy: the CCyB is the **Dutch-specific lever** — and the insurance against a rate shock gone wrong.

Source: ECB Macprudential Bulletin 31 (Detken, Hempell & Pirovano, 2025); DNB Financial Stability Report, spring 2022.

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18/24

SAY

- Thank you. First wider implication: monetary policy.
- The worry was a **double squeeze**, the ECB raising rates while DNB raised capital requirements.
- But Dutch lending rates did not diverge from their peers, and bank profits rose with rates.
- The ECB argues that early buffers “help monetary policy focus on its primary objective of price stability”. Macroprudential policy looks after the financial system so that monetary policy can fight inflation.
- For a euro-area country this matters even more: one monetary policy for twenty countries, but national macroprudential policy. The CCyB is the Dutch-specific lever, and the insurance if a rate shock goes wrong.

NUMBERS

- ECB hiking from Jul 2022 (first +50bp) to at least Sep 2023

TERMS EXPLAINED

- **Macroprudential policy** — policy aimed at the stability of the financial system as a whole
- **Monetary policy** — interest-rate policy aimed at price stability (ECB, euro area-wide)

Through reciprocity every bank lending in the Netherlands holds the 2% — and Europe is moving the same way

Q3 · IMPLICATIONS

Across borders

- Foreign banks with Dutch exposures must also hold the 2% (mandatory reciprocity up to 2.5%)
- Limits leakage to foreign lenders — a known problem with national capital tools (Aiyar, Calomiris & Wieladek, 2014); *non-bank* leakage remains
- In April 2023, 5 banking-union countries had positive-neutral frameworks; NL had the **highest** rate (2%)

CCyB increases in the euro area

Country	Rate	Applies from
Germany	0.75%	Feb 2023
France	1.0%	Jan 2024
Ireland	1.5%	Jun 2024
Netherlands	2.0%	May 2024
Belgium	1.0%	Oct 2024
Portugal	0.75%	Jan 2026
Spain	1.0%	Oct 2026

Source: DNB press release 31 May 2023; ESRB CCyB rates table (Sep 2026); ECB Macroeprudential Bulletin 21 (Behn et al., 2023); Aiyar, Calomiris & Wieladek (2014), JMCB.

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19/24

SAY

- Under EU reciprocity, any foreign bank lending into the Netherlands must also hold the Dutch two percent on those loans. That limits leakage, lending simply moving to foreign banks, a known weakness of national capital rules. Lending can still leak to non-banks.
- And the Netherlands is part of a wider shift, shown in the table: Germany, France, Ireland and Belgium raised their buffers in 2023–24; Portugal and Spain follow in 2026.
- In 2023, five banking-union countries had positive-neutral frameworks, and the Netherlands had the highest rate.

NUMBERS

- Germany 0.75% (Feb 2023) · France 1.0% (Jan 2024) · Ireland 1.5% (Jun 2024) · NL 2.0% (May 2024) · Belgium 1.0% (Oct 2024) · Portugal 0.75% (Jan 2026) · Spain 1.0% (Oct 2026)
- Positive-neutral rates in 2023: Cyprus 0.5%, Estonia and Lithuania 1.0%, Ireland 1.5%, NL 2.0%

TERMS EXPLAINED

- **Reciprocity** — foreign banks apply the host country's CCyB rate to exposures there (mandatory in the EU up to 2.5%)
- **Leakage** — credit shifting to lenders not covered by the rule (foreign branches, non-banks)

Next time, DNB can release €6.7bn overnight

Q3 · IMPLICATIONS

€0 → €6.7bn

releasable-by-design capital

2020 → 2024

≈ half

of the €12bn peak losses

after the GFC

€150bn

lending headroom

if released

Asymmetric payoff (Lang & Menno, 2023)**Normal times:** +1pp requirement → about **−0.1%** lending.**Bad times:** 1pp release → up to about **+10%** lending.

Cheap to hold, powerful to release.

A growing role

The mortgage risk-weight floor expires on 30 Nov 2026.

DNB: this “underpins the importance of . . . the current countercyclical capital buffer (CCyB) of 2%.”

Source: DNB framework (2022); Lang & Menno (2023), ECB WP 2828; DNB press release 24 Apr 2026.

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20/24

SAY

- In 2020 the Netherlands had no capital that was releasable by design. Today it has about 6.7 billion euros, roughly half of the peak losses after the financial crisis, enough to support up to 150 billion of lending.
- ECB research shows why that is a good deal: raising requirements in normal times costs very little lending, about 0.1 percent per point, while a release in bad times can boost lending by up to ten percent.
- And its role is growing. The mortgage risk-weight floor expires at the end of November 2026, and DNB says this “underpins the importance” of the two-percent buffer.

NUMBERS

- €0 → €6.7bn releasable · ≈ half of €12bn peak losses · up to €150bn of lending
- Lang & Menno (2023): **−0.1%** lending per +1pp (normal times) vs up to **+10%** per 1pp release (bad times)

TERMS EXPLAINED

- **State-dependent effect** — the impact of a policy depends on the state of the economy (small in good times, large in bad times)

The price of discretion is credibility — and the buffer has never been tested

Q3 · IMPLICATIONS

Credibility	Without a mechanical rule, markets must trust DNB's judgement and communication — the classic rules-vs-discretion trade-off (Kydland & Prescott, 1977).
Calibration	Why 2%, not 1.5%? The ECB's loss-based range is 1.1–1.8%; NL chose the top of the banking union.
Untested	No release yet: will banks lend the freed capital — or hoard it, as with usable buffers in COVID?
Baseline	Relative to 2021 (not 2019) it <i>was</i> an increase: +€3.3bn before the O-SII offset.
Same profits, two claims	The ECB lists the Netherlands among countries adopting bank levies on excess profits in 2022–23 — partly claiming the earnings that fund the buffer.

Source: Kydland & Prescott (1977); De Nora et al. (2025); ECB Macroprudential Bulletin 21 (2023) and 31 (2025); DNB (2022, 2023).

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21/24

SAY

- **Credibility.** Without a mechanical rule, markets must trust DNB's judgement. That is the classic rules-versus-discretion trade-off: a rule is predictable but can be wrong, as we saw; discretion must be explained every quarter.
- **Calibration.** Why two percent? ECB estimates suggest 1.1 to 1.8.
- **Untested.** We do not yet know whether banks will lend released capital or hoard it.
- **Baseline.** Measured against 2021 rather than 2019, it *was* an increase of 3.3 billion before the O-SII offset.
- **Same profits, two claims.** The ECB lists the Netherlands among countries that levied excess-profit taxes on banks in 2022–23, partly the same earnings that fund the buffer.

NUMBERS

- ECB loss-based range 1.1–1.8% vs NL 2% · +€3.3bn vs 2021 before the O-SII offset

TERMS EXPLAINED

- **Rules vs discretion** — the trade-off between predictable formulas and flexible judgement (Kydland & Prescott 1977)
- **Excess-profit levy** — a temporary tax on bank profits seen as windfalls from higher rates

A promise kept: the buffer was refilled in the storm — at the moment it was cheapest

EPILOGUE

Q1 · Environment

Recovered economy hit by war, inflation and rate hikes; credit cold; housing hot, then cooling; strong, profitable banks.

Q2 · Criteria & risks

2% normal-times level, calibrated on past losses; insurance against unforeseeable shocks; housing to other tools; structural buffers cut in return.

Q3 · Implications

Cheap · largely a swap · no sign of contraction · complements monetary policy · reciprocated abroad · €6.7bn releasable · relies on discretion.

In March 2020, the Netherlands had nothing to release.
Next time, it will.

The best time to fill a buffer is when nobody thinks you need it — even in a storm.

Huang & Gutiérrez · The Netherlands' CCyB

22/24

SAY

- **Environment:** a recovered economy hit by war, inflation and rate hikes; credit cold; housing hot, then cooling; strong, profitable banks.
- **Criteria and risks:** a two-percent normal-times level calibrated on past losses, insuring against shocks no one can forecast. Housing went to other tools; structural buffers were cut in return.
- **Implications:** cheap, largely a swap, no sign of contraction, complementary to monetary policy, reciprocated abroad, and 6.7 billion euros now releasable. The price is reliance on discretion.
- The one idea to take away: this was less about *how much* capital banks hold and more about *which kind*.
- *(pause)* In March 2020, the Netherlands had nothing to release. **Next time, it will.** The best time to fill a buffer is when nobody thinks you need it, even in a storm. Thank you; we look forward to your questions.

NUMBERS

- 0% → 2% · −33 / −47pp gap · $\leq 1/5$ of headroom · €6.7bn releasable

TERMS EXPLAINED

- **Normalisation** — returning the buffer to its normal-times level, not reacting to overheating

Glossary

APPENDIX

CCyB	Countercyclical capital buffer: 0–2.5% of risk-weighted assets on domestic exposures; raised with a 12-month lag, released immediately.	Credit-to-GDP gap	Credit/GDP ratio minus its long-run trend; the Basel reference for the CCyB (gap 2–10pp → buffer 0–2.5%).
Positive neutral CCyB	A CCyB kept above zero when risks are neither high nor low, so it can be released for any shock.	HP filter	Hodrick-Prescott filter: estimates a smooth trend; one-sided uses only past data (real time), two-sided also future data.
CET1	Common Equity Tier 1: the highest-quality capital (shares, retained earnings), as % of RWA.	Headroom	CET1 capital above a bank's requirement / MDA trigger.
RWA	Risk-weighted assets: assets weighted by their riskiness (e.g. mortgages count less than corporate loans).	Reciprocity	Foreign banks must apply the Dutch CCyB rate to their Dutch exposures.
Pillar 1 / P2R / P2G	Legal minimum (4.5% CET1) / bank-specific add-on set by the supervisor / non-binding supervisory guidance.	Borrower-based measures	Limits on loan-to-value (LTV) and loan-to-income (LTI) of new mortgages.
CCoB	Capital conservation buffer, 2.5% for all banks.	Risk-weight floor	Minimum average risk weight on Dutch mortgages of banks using internal models (Art. 458 CRR), Jan 2022 – Nov 2026.
O-SII / SRB	Buffer for other systemically important institutions / systemic risk buffer (structural, not designed for release).	Event study / placebo	Compare NL with countries that did not change their CCyB around each date / repeat with each control as a fake "treated" country.
MDA	Maximum distributable amount: if CET1 falls below the combined buffer requirement, payouts are automatically restricted.	pp / bp	Percentage point / basis point (0.01pp).

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23/24

SAY

- Backup only. Use it when a question involves a term; point to the row and give the one-line definition.

NUMBERS

- (definitions only)

TERMS EXPLAINED

- All key terms of the talk are on the slide

Methods and references

APPENDIX

Methods. Credit gap: one-sided (recursive) and two-sided HP filter, $\lambda = 400,000$, sample from 1961Q1; Hamilton (2018) regression filter, $h = 8, 20, p = 4$. Headroom: (CET1 ratio – CET1 requirement/MDA) $\times$ total RWA per bank; CCyB amount sector-wide (upper bound); 6–47% assumption grid as sensitivity. Event study: two-way FE, NL $\times$ 3-month event-time bins, country and month FE; placebo inference over controls with no CCyB change 2021–25. Code and data: GitHub repository (code/, data/).

Primary sources

BCBS (2024). Range of practices in implementing a positive neutral CCyB (d585).
 DNB (2020). DNB lowers bank buffer requirements to support lending, 17 Mar.
 DNB (2022). Analytical framework for setting the CCyB in the Netherlands.
 DNB (2022, 2023). CCyB press releases, 27 May 2022, 31 May 2023; O-SII news item, 31 May 2023.
 DNB (2026). CCyB decisions 26 Mar and 17 Sep; risk-weight measure expires, 24 Apr.
 DNB. Financial Stability Report, spring 2022 and spring 2023.
 Knot, K. (2023). Introductory statement, Standing Parliamentary Committee for Finance, 7 Jun.
 ECB (2020). Temporary capital and operational relief, 12 Mar; ECB (2022). Monetary policy decisions, 21 Jul.
 ESRB. CCyB rates table; O-SII notifications NL (27 Nov 2020; 31 May 2023); COVID-19 measures.
 Data: BIS (credit gaps, total credit, DSR, property prices); ECB Data Portal (MIR, BSI); CBS; bank annual and Pillar 3 reports (ING, Rabobank, ABN AMRO, de Volksbank).

Literature

Aiyar, Calomiris & Wieladek (2014). Does macro-prudential regulation leak? *JMCB* 46(s1).
 Behn, Pereira, Pirovano & Testa (2023). A positive neutral rate for the CCyB. *ECB MPB* 21.
 Couaillier, Lo Duca, Reghezza & Rodriguez d'Acri (2022). Caution: do not cross! *ECB WP* 2644.
 De Nora, Pereira, Pirovano & Stammwitz (2025). From losses to buffer. *ECB WP* 3061.
 Detken, Hempell & Pirovano (2025). Macroprudential and monetary policy interaction. *ECB MPB* 31.
 Drehmann, Borio, Gambacorta, Jiménez & Trucharte (2010). Countercyclical capital buffers: exploring options. *BIS WP* 317.
 Edge & Meisenzahl (2011). The unreliability of credit-to-GDP ratio gaps in real time. *IJCB* 7(4).
 Hamilton (2018). Why you should never use the Hodrick-Prescott filter. *REStat* 100(5).
 Herrera, Scalone & Pirovano (2024). The importance of being positive. *ECB MPB* 24.
 Kydland & Prescott (1977). Rules rather than discretion. *JPE* 85(3).
 Lang & Menno (2023). The state-dependent impact of changes in bank capital requirements. *ECB WP* 2828.
 Repullo & Saurina (2011). The countercyclical capital buffer of Basel III: a critical assessment. CEMFI WP 1102.
 Tinbergen (1952). *On the Theory of Economic Policy*. North-Holland.

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24/24

SAY

- Backup for method questions.
 - **Credit gap:** one-sided and two-sided HP filter ($\lambda = 400,000$), plus Hamilton regression filter.
 - **Headroom:** (CET1 ratio minus requirement) $\times$ total RWA, per bank.
 - **Event study:** two-way fixed effects, three-month bins, placebo inference over five never-treated countries.
- All references are verified.

NUMBERS

- Data: BIS, ECB Data Portal (MIR, BSI), ESRB, DNB, bank reports

TERMS EXPLAINED

- **Two-way fixed effects** — regression with country and time dummies, so differences are measured relative to common trends